

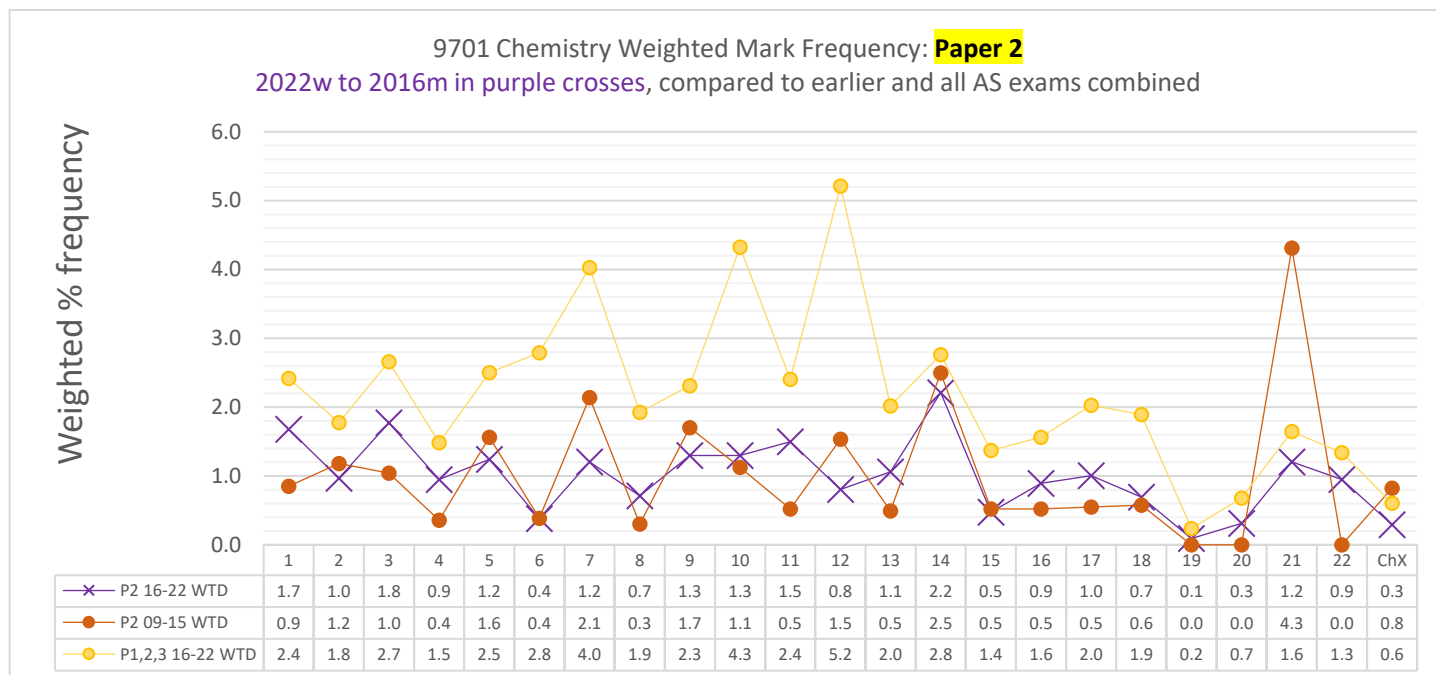
ALvl Chem 9 EQ P2 22w to 09s Paper 2 The Periodic Table: chemical periodicity 133marks

As you start and work through this worksheet you can tick off your progress to show yourself how much you have done, and what you need to do next. The first task is just to read the first question and should take you less than one minutes to complete.

Paper 2 Topic 9

Checklist Tick each task off as you go along

RANK:		P1 Noob	P1 Novice	P1 Bronze	P1 Silver	P1 Gold	P1 ¹ Winner	P1 Hero	P1 Legend
		1 Q started	1 Q done	10% of marks	25% of marks	40% of marks	50% of marks	75% of marks	100% of marks
Topic (marks)	133		5	13	33	53	67	100	133
Time @75s/mark (minutes)	166		6	17	42	67	83	125	166



What the most thoughtful students will get out of their extensive studying will be a capacity to do meaningful brain-based work even under stressful conditions, which is a part of the self-mastery skillset that will continue to deliver value for the whole of their lives. Outstanding grades will also happen, but the most important goal from skillful action in study is being better at any important task, even if circumstances do not feel ideal.

As you are moving through your studies you can learn more about yourself by trying out new ways to manage yourself, and analysing how effective those new techniques were. In this reflective process not only will you get better at working positively and productively to deliver ambitious and successful outcomes, but you will be working towards one aspect of life's highest pursuit, summarised and inscribed on the Temple of Apollo at Delphi: "know thyself".

1. To complete these questions, as important as your answer, is checking your answer against the mark scheme.
2. For each page or group of 10-20 marks, convert your mark score into a percentage. This will allow you to see (and feel) your progress as you get more experience and understanding with each topic.
3. Multiple choice questions, done carefully where you explain and show yourself your thinking using written notes as you move through each question, can be more useful than just Paper 2 for students aiming for a C or B grade. Paper 2 should be the larger focus for students aiming for A and A* grades, however.
4. If you find you get a higher percentage answering short answer questions than multiple choice questions that often means you are NOT using the marking scheme correctly; your correct answer might not be fully complete for all the marks you are awarding. The marks easiest to miss rely on providing the largest amount of detail.

¹ **DO NOT** work on these higher levels of completion in your AS year unless you have also achieved at least a "Silver" (25%) in the same topic in **Paper 1**, which tend also to be easier questions, as well as "Silver" (25%) in the same topic, if it exists, in Paper 3.

Inorganic chemistry

9 The Periodic Table: chemical periodicity

9.1 Periodicity of physical properties of the elements in Period 3

Learning outcomes

Candidates should be able to:

- 1 describe qualitatively (and indicate the periodicity in) the variations in atomic radius, ionic radius, melting point and electrical conductivity of the elements
- 2 explain the variation in melting point and electrical conductivity in terms of the structure and bonding of the elements

9.2 Periodicity of chemical properties of the elements in Period 3

Learning outcomes

Candidates should be able to:

- 1 describe, and write equations for, the reactions of the elements with oxygen (to give Na_2O , MgO , Al_2O_3 , P_4O_{10} , SO_2), chlorine (to give NaCl , MgCl_2 , AlCl_3 , SiCl_4 , PCl_5) and water (Na and Mg only)
- 2 state and explain the variation in the oxidation number of the oxides (Na_2O , MgO , Al_2O_3 , P_4O_{10} , SO_2 and SO_3 only) and chlorides (NaCl , MgCl_2 , AlCl_3 , SiCl_4 , PCl_5 only) in terms of their outer shell (valence shell) electrons
- 3 describe, and write equations for, the reactions, if any, of the oxides Na_2O , MgO , Al_2O_3 , SiO_2 , P_4O_{10} , SO_2 and SO_3 with water including the likely pHs of the solutions obtained
- 4 describe, explain, and write equations for, the acid/base behaviour of the oxides Na_2O , MgO , Al_2O_3 , P_4O_{10} , SO_2 and SO_3 and the hydroxides NaOH , Mg(OH)_2 , Al(OH)_3 including, where relevant, amphoteric behaviour in reactions with acids and bases (sodium hydroxide only)
- 5 describe, explain, and write equations for, the reactions of the chlorides NaCl , MgCl_2 , AlCl_3 , SiCl_4 , PCl_5 with water including the likely pHs of the solutions obtained
- 6 explain the variations and trends in 9.2.2, 9.2.3, 9.2.4 and 9.2.5 in terms of bonding and electronegativity
- 7 suggest the types of chemical bonding present in the chlorides and oxides from observations of their chemical and physical properties

9.3 Chemical periodicity of other elements

Learning outcomes

Candidates should be able to:

- 1 predict the characteristic properties of an element in a given group by using knowledge of chemical periodicity
- 2 deduce the nature, possible position in the Periodic Table and identity of unknown elements from given information about physical and chemical properties

3 Some of the common chlorides of Period 3 elements are shown in the list.



(a) From this list, identify:

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

..... [1]

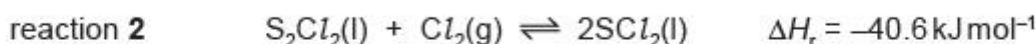
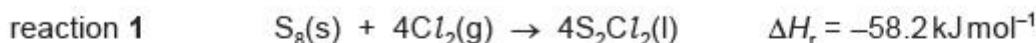
(iii) the chloride that dissolves in water to form a neutral solution

..... [1]

(iv) the chloride formed from the **element** with the highest melting point.

..... [1]

(d) Sulfur, S_8 , reacts with chlorine to form several different chlorides. The most common are S_2Cl_2 and SCl_2 . SCl_2 forms when sulfur reacts with an excess of chlorine.



(i) SCl_2 is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl_2 .
Explain your answer.

.....
.....
..... [2]

Q# 183/ ALvl Chemistry/2022/s/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

(d) Separate samples of **Q** and **R** are added to separate test-tubes containing acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ and heated.

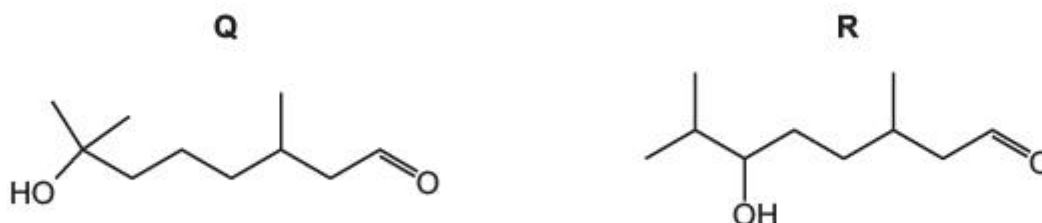


Fig. 3.3

(ii) When $\text{PCl}_5(\text{s})$ is added to separate samples of **Q** and **R** at room temperature, both react vigorously.

- (iii) Suggest why samples of **Q** and **R** must be dried before PCl_5 is added. Include a relevant equation to support your answer.

.....
.....
..... [2]

[Total: 17]

Q# 184/ ALvl Chemistry/2022/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

- 1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.
- (f) Magnesium, Mg, burns in oxygen, O_2 .
The activation energy, E_a , for this reaction is $+148 \text{ kJ mol}^{-1}$.
- (i) State **one** observation when magnesium burns in oxygen.
Do **not** refer to temperature changes in your answer.

..... [1]

Q# 185/ ALvl Chemistry/2022/m/TZ 2/Paper 4/Q# 3/www.SmashingScience.org

- 3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.
- (b) HCl is a product of several different reactions. Some of these are shown in Fig. 3.1.

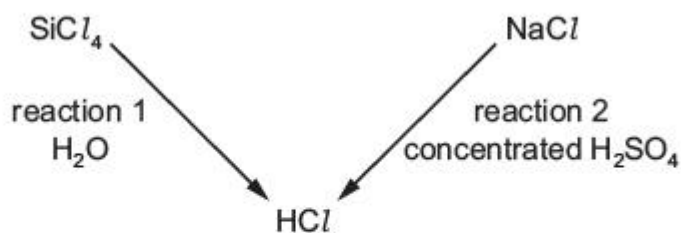


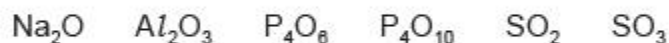
Fig. 3.1

- (i) Write an equation for reaction 1.

..... [1]

Q# 186/ ALvl Chemistry/2022/m/TZ 2/Paper 4/Q# 2/www.SmashingScience.org

- 2 Some oxides of elements in Period 3 are shown.



- (a) Na reacts with O_2 to form Na_2O . Na is the reducing agent in this reaction.

(ii) Write an equation for the reaction of Na_2O with water.

..... [1]

(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(i) State what is meant by amphoteric.

.....
..... [1]

(ii) Al_2O_3 is purified from bauxite in several steps. The first step involves heating Al_2O_3 with an excess of NaOH(aq) . A colourless solution forms.

Write an equation for this reaction.

..... [1]

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(i) Deduce the type of structure and bonding shown by P_4O_6 . Explain your answer.

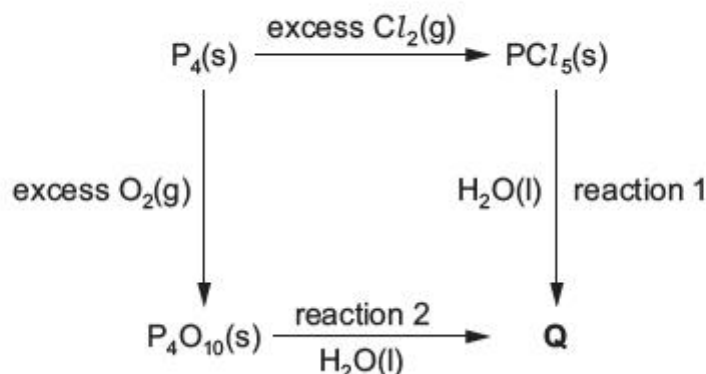
.....
.....
..... [2]

(iv) Write an equation for the reaction of P_4O_{10} with water.

..... [1]

Q# 187/ ALvl Chemistry/2021/w/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

(b) Some reactions of $\text{P}_4(\text{s})$ are shown in the reaction scheme.



(ii) Deduce the identity of **Q** and hence construct chemical equations for reactions 1 and 2.

reaction 1 $\text{PCl}_5 + \dots \text{H}_2\text{O} \rightarrow \dots$

reaction 2 $\text{P}_4\text{O}_{10} + \dots \text{H}_2\text{O} \rightarrow \dots$

[2]

(e) Aluminium reacts with chlorine to form aluminium chloride.

Aluminium chloride can exist as the gaseous molecule $Al_2Cl_6(g)$. This molecule contains coordinate bonds.

(ii) Describe what you would see when solid aluminium chloride reacts with water.

Name the type of reaction that occurs.

.....
.....
..... [2]

(f) 0.020 mol of element **Z** reacts with excess Cl_2 to form 0.020 mol of a liquid chloride.

The liquid chloride has formula ZCl_n , where n is an integer.

ZCl_n reacts vigorously with water at room temperature to give an acidic solution and a white solid.

When excess $AgNO_3(aq)$ is added to the solution, 11.54 g of $AgCl(s)$ forms.

(i) Suggest the type of bonding and structure shown by ZCl_n .

..... [1]

(ii) Calculate the value of n in ZCl_n .

$n =$ [2]

3 The reducing agent $LiAlH_4$ can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

(b) $LiAlH_4$ cannot be used in aqueous solution because it reacts with water to produce $LiOH(aq)$, $H_2(g)$ and a white precipitate which is soluble in excess sodium hydroxide.

Identify the white precipitate.

..... [1]



2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(a) Phosphorus reacts with an excess of oxygen to form phosphorus(V) oxide.

(i) Write an equation to show the reaction of phosphorus with excess oxygen.

..... [1]

(ii) Describe the reaction of phosphorus(V) oxide with water.

.....
.....
..... [2]

(b)

(ii) Write the equation for the reaction of SO_2 with water.

..... [1]

(e) Element **E** is a Period 5 element.

E reacts with oxygen to form an insoluble white oxide that has a melting point of 1910°C . The oxide of **E** conducts electricity only when liquid.

E also reacts readily with $\text{Cl}_2(\text{g})$ to form a white solid that reacts exothermically with water. The resulting solution reacts with aqueous silver nitrate to form a white precipitate that dissolves in dilute ammonia.

(i) Suggest the type of bonding shown by the **oxide** of **E**. Explain your answer.

.....
.....
..... [2]

(ii) Suggest the type of bonding shown by the **chloride** of **E**. Explain your answer.

.....
.....
..... [2]

[Total: 21]

1 Gallium is a metal in Group 13 of the Periodic Table.

(a) There are two stable isotopes of gallium, ^{69}Ga and ^{71}Ga .



(c) Gallium metal reacts rapidly when exposed to air. A white solid layer is formed on its surface.

- (i) Suggest an equation to describe the reaction occurring when gallium metal is exposed to air.

..... [2]

- (ii) The table gives the formula of each gallium-containing product formed when gallium oxide reacts separately with hot aqueous hydrochloric acid and hot aqueous sodium hydroxide.

	formula of gallium-containing product
hot aqueous hydrochloric acid	GaCl_3
hot aqueous sodium hydroxide	NaGa(OH)_4

Give the name of the type of behaviour shown by gallium oxide in these reactions.

..... [1]

[Total: 8]

Q# 192/ ALvl Chemistry/2020/m/TZ 2/Paper 4/Q# 1/www.SmashingScience.org

(e) Unlike the other oxides of Group 2 metals, beryllium oxide is amphoteric.

- (i) Give the meaning of the term *amphoteric*.

.....

..... [1]

- (ii) Beryllium oxide and aluminium oxide have similar chemical properties.

The Be(OH)_4^{2-} anion is a product of the reaction between beryllium oxide and excess concentrated $\text{OH}^-(\text{aq})$.

Construct an equation for this reaction.

..... [1]

Q# 193/ ALvl Chemistry/2019/w/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

	Na_2O	MgO	Al_2O_3	SiO_2	SO_3
boiling point/ $^\circ\text{C}$	1275	3670	2977	2950	45
nature of oxide	basic	basic	amphoteric	acidic	acidic

(b)

(ii) Al_2O_3 is an amphoteric oxide.

Explain what is meant by the term *amphoteric*. Use chemical equations to illustrate your answer.

.....
.....
..... [3]

(iii) State what you would observe when a small sample of Na_2O is placed in water.

..... [1]

(c)

(iii) SeO_2 shows similar chemical reactions to SO_2 .

Suggest an equation to show the reaction of SeO_2 with aqueous sodium hydroxide, $NaOH$.

..... [1]

Q# 194/ ALvl Chemistry/2019/s/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

(f) Silicon dioxide reacts with hot, concentrated sodium hydroxide.

(i) Identify the **two** products formed during this reaction.

.....
..... [2]

(ii) Describe the behaviour of the silicon dioxide during this reaction.

..... [1]

Q# 195/ ALvl Chemistry/2018/s/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

(c) The elements in the third period, from sodium to silicon, can react with chlorine to form chlorides.

(i) State and explain the pattern of change of oxidation number which occurs to both chlorine and the different Period 3 elements when they react together.

.....
.....
.....
..... [3]

(ii) Give the equations to show the reactions of sodium chloride and silicon(IV) chloride when separately added to water.

sodium chloride

silicon(IV) chloride

- (iii) Complete the table to describe the structure and bonding in sodium chloride and silicon(IV) chloride.

	structure	bonding
sodium chloride		
silicon(IV) chloride		

[2]

[Total: 16]

Q# 196/ ALVI Chemistry/2017/w/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

- 2 The elements in the third period, and their compounds, show trends in their physical and chemical properties.
- (b) The chlorides of the elements in the third period behave in different ways when added to water, depending on their structure and bonding.

L and **M** are each a chloride of an element in Period 3. A student investigated **L** and **M** and their results are given.

L is a white crystalline solid with a melting point of 987K. **L** dissolves in water to form an approximately neutral solution. Addition of NaOH(aq) to an aqueous solution of **L** produces a white precipitate.

M is a liquid with a boiling point of 331K. **M** is hydrolysed rapidly by cold water to form a strongly acidic solution, a white solid and white fumes.

Identify **L** and **M**.

Explain any properties and observations described.

Give equations where appropriate.

- (i) **L** is
-
-
-
- [3]

- (ii) **M** is
-
-
-
- [3]

[Total: 14]

[illegible]

(i) the element that forms an amphoteric oxide, [1]

(ii) the element with the highest first ionisation energy, [1]

(iii) the element that forms a soluble hydroxide and an insoluble sulfate, [1]

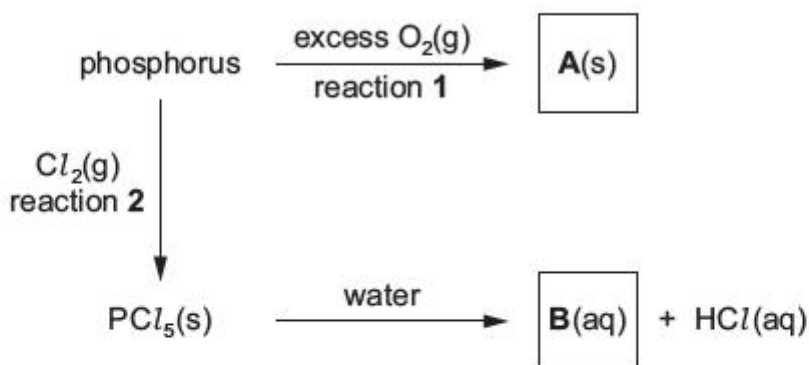
(iv) the most volatile element in a group that contains elements in all three states of matter at room temperature and pressure, [1]

(v) the element that forms the largest cation. [1]

element	Na	Mg	Al	P	S	Cl
atomic radius/nm	0.186	0.160	0.143	0.110	0.104	0.099
radius of most common ion/nm	0.095	0.065	0.050	0.212	0.184	0.181
maximum oxidation number of the element in its compounds	+1					+7

- (b) Phosphorus is a non-metal in the third period. It reacts vigorously with excess oxygen but slowly with chlorine.

Some reactions of phosphorus are shown.



- (i) Write an equation to represent reaction 1, the formation of compound A.

..... [1]

- (ii) Give **two** observations you could make in reaction 2.

1.

2.

[2]

- (iii) Name compound B.

..... [1]

Q# 199/ ALvl Chemistry/2016/s/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

- 2 D, E, F, and G are four consecutive elements in the **fourth** period of the Periodic Table. (The letters are **not** the actual symbols of the elements.)

D is a soft, silvery metal with a melting point just above room temperature. Its amphoteric oxide, D_2O_3 , has a melting point of 1900°C and can be formed by heating D in oxygen.

G is a solid that can exist as several different allotropes, most of which contain G_8 molecules. G burns in air to form GO_2 which dissolves in water to form an acidic solution. This solution reacts with sodium hydroxide to form the salt Na_2GO_3 .

- (a) Suggest the identities of D and G.

D G

[1]

- (b) Write equations for the reactions of D_2O_3 with

- (i) hydrochloric acid,

..... [2]

(ii) sodium hydroxide.

..... [2]

(c) Suggest the type of bonding and structure in D_2O_3 .

..... [1]

(d) Write an equation for the formation of an acidic solution when GO_2 dissolves in water.

..... [1]

Q# 200/ ALvl Chemistry/2015/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

(c) The element tellurium, Te, reacts with chlorine to form a single solid product, with a relative formula mass of 270. The product contains 52.6% chlorine by mass.

(ii) This chloride melts at 224°C and reacts vigorously with water.

State the type of bonding **and** structure present in this chloride and explain your reasoning.

.....
.....
.....
..... [2]

(iii) Suggest an equation for the reaction of this chloride with water.

..... [1]

(d) Sodium and silicon also react directly with chlorine to produce the chlorides shown.

chloride	melting point/ $^\circ\text{C}$	difference between the electronegativities of the elements
NaCl	801	2.2
SiCl_4	-69	1.3

(i) Describe what you would **see** during the reaction between sodium and chlorine.

.....
.....
..... [2]



- (ii) Explain the differences between the melting points of these two chlorides in terms of their structure **and** bonding. You should refer to the difference between the electronegativities of the elements in your answer.

NaCl structure **and** bonding

.....

SiCl₄ structure **and** bonding

.....

explanation

.....

.....

.....

.....

..... [4]

[Total: 20]

Q# 201/ ALvl Chemistry/2014/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

- (d) The chlorides of elements in Period 3 of the Periodic Table show different behaviours on addition to water, depending on their structure and bonding.

- (i) Write equations to show the behaviour of sodium chloride, NaCl, and silicon chloride, SiCl₄, when separately added to an excess of water.

NaCl

SiCl₄

[2]

- (ii) State and explain the differences in behaviour of these two chlorides when added to water, in terms of their structure and the bonding found in the compounds.

.....

.....

.....

.....

.....

..... [4]

- (b) Chlorine is very reactive and will form compounds by direct combination with many elements.

Describe what you would see when chlorine is passed over separate heated samples of sodium and phosphorus.

In **each** case write an equation for the reaction.

sodium

.....

.....

.....

phosphorus

.....

.....

..... [4]

- (d) Magnesium chloride, MgCl_2 , and silicon tetrachloride, SiCl_4 , each dissolve in or react with water.

Suggest the approximate pH of the solution formed in **each** case.

MgCl_2 SiCl_4

Explain, with the aid of an equation, the difference between the two values.

.....

.....

.....

..... [5]

- 3 This question refers to the elements in the section of the Periodic Table shown below.

		H						He		
Li	Be			B	C	N	O	F	Ne	
Na	Mg			Al	Si	P	S	Cl	Ar	
K	Ca		transition elements		Ga	Ge	As	Se	Br	Kr



(a) From this list of elements, identify in **each** case **one** element that has the property described. Give the **symbol** of the element.

(i) An element that floats on cold water and reacts readily with it.

.....

(ii) An element that forms an oxide that is a reducing agent.

.....

(iii) The element that has the smallest first ionisation energy.

.....

(iv) The element which has a giant molecular structure **and** forms an oxide which has a simple molecular structure.

.....

(v) The element in Period 3 (Na to Ar) that has the smallest anion.

.....

(vi) The element in Period 3 (Na to Ar) which forms a chloride with a low melting point and an oxide with a very high melting point.

.....

[6]

(b) Use the elements in Period 3 (Na to Ar) in the section of the Periodic Table opposite to identify the oxide(s) referred to below.

In **each** case, give the **formula** of the oxide(s).

(i) An oxide which when placed in water for a long time has no reaction with it.

.....

(ii) An oxide which dissolves readily in water to give a strongly alkaline solution.

.....

(iii) Two acidic oxides formed by the same element.

..... and

(iv) An oxide which is amphoteric.

.....

[5]

- 1 Zinc is an essential trace element which is necessary for the healthy growth of animals and plants. Zinc deficiency in humans can be easily treated by using zinc salts as dietary supplements.

- (a) One salt which is used as a dietary supplement is a hydrated zinc sulfate, $\text{ZnSO}_4 \cdot x\text{H}_2\text{O}$, which is a colourless crystalline solid.

Crystals of zinc sulfate may be prepared in a school or college laboratory by reacting dilute sulfuric acid with a suitable compound of zinc.

Give the formulae of **two** simple compounds of zinc that could **each** react with dilute sulfuric acid to produce zinc sulfate.

..... and [2]

- 1 Oxides are compounds which usually contain oxygen combined with one other element.

Oxides are classified as follows.

acidic alkaline amphoteric basic

- (a) Using these terms only, complete the table to describe the oxides of the elements of the third period of the Periodic Table sodium to sulfur.

Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2	Cl_2O_7
						acidic

[4]

- (b) Give the names of **two** elements from sodium to chlorine which form more than one oxide.

..... and [1]

- (c) Sodium reacts with water.

- (i) Describe, as fully as you can, what you would see when a piece of sodium is reacted with water.

.....

- (ii) Write an equation for the reaction of sodium with water.

.....

[4]

element	Na	Mg	Al	Si	P	S
conductivity			high			
melting point			high			

- (d) (i) Complete the 'conductivity' row by using **only** the words 'high', 'moderate' or 'low'.
(ii) Complete the 'melting point' row by using **only** the words 'high' or 'low'. [5]

When Mendeleev published his first Periodic Table, he left gaps for elements that had yet to be discovered. He also predicted some of the physical and chemical properties of these undiscovered elements.

For one element, **E**, he correctly predicted the following properties.

melting point of the element	high
------------------------------	------

melting point of the oxide	high
----------------------------	------

boiling point of the chloride	low
-------------------------------	-----

The element **E** was in the fourth Period and was one of the elements from gallium, proton number 31, to bromine, proton number 35.

- (e) By considering the properties of the third Period elements aluminium to chlorine, suggest the identity of the fourth Period element **E**.

[1]

																H																	He
Li	Be															B	C	N	O	F	Ne												
Na	Mg															Al	Si	P	S	Cl	Ar												
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr																

- (a) From this table, identify in **each** case **one** element that has the property described. Give the **symbol** of the element in each case.

- (i) The element that has a molecule which contains exactly eight atoms.

.....

- (ii) The element that forms the largest cation.

(iii) An element that floats on water and reacts with it.

.....

(iv) An element that reacts with water to give a solution that can behave as an oxidising agent.

.....

(v) An element whose nitrate gives a brown gas on thermal decomposition.

.....

[5]

(b) (i) Give the formula of the oxide of the most electronegative element.

.....

(ii) Several of these elements form more than one acidic oxide.
Give the formulae of **two** such oxides formed by the **same** element.

..... and

[3]

Q# 208/ ALvI Chemistry/2009/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

1 Copper and titanium are each used with aluminium to make alloys which are light, strong and resistant to corrosion.

Aluminium, Al , is in the third period of the Periodic Table; copper and titanium are both transition elements.

Aluminium reacts with chlorine.

Copper forms two chlorides, $CuCl$ and $CuCl_2$.

(e) At room temperature, the chloride of titanium, A , is a liquid which does not conduct electricity.

What does this information suggest about the bonding and structure in A ?

.....

.....

..... [2]

[Total: 14]

Q# 209/ ALvI Chemistry/2009/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

1 Copper and titanium are each used with aluminium to make alloys which are light, strong and resistant to corrosion.

Aluminium, Al , is in the third period of the Periodic Table; copper and titanium are both transition elements.

Aluminium reacts with chlorine.

- (b) (i) Outline how, starting from aluminium powder, this reaction could be carried out in a school or college laboratory to give a small sample of aluminium chloride. A diagram is not necessary.

.....

.....

.....

- (ii) Describe what you would see during this reaction.

.....

.....

[4]

Mark Scheme ALvI Chem 9 EQ P2 22w to 09s Paper 2 The Periodic Table: chemical periodicity 133marks

Q# 182/ ALvI Chemistry/2022/w/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

3(a)(ii)	AlCl ₃ AND SiCl ₄ AND PCl ₅	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl ₄	1
3(d)(i)	M1 (structure =) simple / molecular, because it has a low melting / boiling point	1
	M2 (bonding =) covalent, because it is hydrolysed	1

Q# 183/ ALvI Chemistry/2022/s/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

3(d)(iii)	M1 water reacts with / hydrolyses PCl ₅ M2 H ₂ O + PCl ₅ → POCl ₃ + 2HCl OR 4H ₂ O + PCl ₅ → H ₃ PO ₄ + 5HCl	2
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Q# 184/ ALvI Chemistry/2022/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

1(f)(i)	white flame / light OR white solid / smoke	1
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Q# 185/ ALvI Chemistry/2022/m/TZ 2/Paper 4/Q# 3/www.SmashingScience.org

3(b)(i)	SiCl ₄ + 2H ₂ O → SiO ₂ + 4HCl	1
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Q# 186/ ALvI Chemistry/2022/m/TZ 2/Paper 4/Q# 2/www.SmashingScience.org

2(a)(ii)	Na ₂ O + H ₂ O → 2NaOH	1
2(b)(i)	reacts with both acids and bases / shows both acidic and basic behaviour	1
2(b)(ii)	Al ₂ O ₃ + 2NaOH + 3H ₂ O → 2NaAl(OH) ₄	1
2(c)(i)	(structure =) simple/molecular, because it has a low melting/boiling point [1] (bonding =) covalent, because it is hydrolysed [1]	2
2(c)(iv)	P ₄ O ₁₀ + 6H ₂ O → 4H ₃ PO ₄	1

Q# 187/ ALvI Chemistry/2021/w/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

3(b)(ii)	PCl ₅ + 4H ₂ O → H ₃ PO ₄ + 5HCl P ₄ O ₁₀ + 6H ₂ O → 4H ₃ PO ₄	2
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Q# 188/ ALvl Chemistry/2021/m/TZ 2/Paper 4/Q# 2/www.SmashingScience.org

2(e)(ii)	M1: (AlCl₃ / solid) disappears - misty / steamy fumes - temperature increases M2: hydrolysis	2
2(f)(i)	simple / molecular AND covalent	1
2(f)(ii)	M1: $11.54 \div 143.4 = 0.0805$ M2: so ratio Z:Cl is 1:4 / $n = 4$	2

Q# 189/ ALvl Chemistry/2020/w/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

3(b)	Al(OH) ₃ / aluminium hydroxide	1
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Q# 190/ ALvl Chemistry/2020/w/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

2(a)(i)	$P_4 + 5O_2 \rightarrow P_4O_{10}$	1
2(a)(ii)	any two from: - reacts vigorously - solid disappears / colourless solution forms - hydrolysis - exothermic - acid(ic) (solution) - steamy / misty fumes	2
2(b)(ii)	$SO_2 + H_2O \rightarrow H_2SO_3$	1
2(e)(i)	M1: ionic M2: ions only able / free to move / free to conduct (when liquid / molten)	2
2(e)(ii)	M1: covalent M2: hydrolysed (by water)	2

Q# 191/ ALvl Chemistry/2020/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

1(c)(i)	$4Ga + 3O_2 \rightarrow 2Ga_2O_3$ M1 correct formula of Ga ₂ O ₃ M2 correctly balanced equation based on Ga + O ₂ and formula of gallium oxide in M1	2
1(c)(ii)	amphoteric	1

Q# 192/ ALvl Chemistry/2020/m/TZ 2/Paper 4/Q# 1/www.SmashingScience.org

1(e)(i)	reacts with / behaves as both acid and base	1
1(e)(ii)	$BeO + 2OH^- + H_2O \rightarrow Be(OH)_4^{2-}$	1

Q# 193/ ALvl Chemistry/2019/w/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

2(b)(ii)	M1: reacts with both acid and base / alkali M2: use any equation with Al ₂ O ₃ and an acid, e.g. $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ M3: use any equation with Al ₂ O ₃ and a base / alkali, e.g. $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$	3
2(b)(iii)	solid dissolves / disappears OR gets warm / hot	1
2(c)(iii)	$SeO_2 + 2NaOH \rightarrow Na_2SeO_3 + H_2O$	1

Q# 194/ ALvl Chemistry/2019/s/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

2(f)(i)	M1 sodium silicate / Na ₂ SiO ₃ M2 water / H ₂ O	1
2(f)(ii)	acid(ic)	1



Q# 195/ ALvl Chemistry/2018/s/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

3(c)(i)	oxidation numbers / states of elements (Na-Si) increase from +1 to +4 / by 1 every time			1
	increasing number of valence electrons / NaCl, MgCl ₂ , AlCl ₃ , SiCl ₄ / number of chlorines matches group number			1
	chlorine oxidation number / state -1 in all / stays the same			1
3(c)(ii)	NaCl → Na ⁺ + Cl ⁻			1
	SiCl ₄ + 2H ₂ O → SiO ₂ + 4HCl			1
3(c)(iii)		structure	bonding	2
	sodium chloride	giant / ionic	ionic	
	silicon(IV) chloride	simple / molecular	covalent	

Q# 196/ ALvl Chemistry/2017/w/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

2(b)(i)	(L=) MgCl ₂ / magnesium chloride	1
	Any two from (giant) ionic (with strong attractions) Mg²⁺(aq) / Mg(H₂O)₆²⁺(aq) is neutral / undergoes (partial) hydrolysis Mg(OH)₂ is the white precipitate / solid / insoluble / partially soluble MgCl₂ + 2NaOH → Mg(OH)₂ + 2NaCl	2
2(b)(ii)	(M=) SiCl ₄ / silicon chloride	1
	Any two from (simple) molecular / simple covalent hydrolysis possible due to available d orbitals forms HCl (aq) / hydrochloric acid / solution and / or HCl gas / fumes white solid is (hydrated) SiO₂ SiCl₄ + 2H₂O → SiO₂ + 4HCl	2

Q# 197/ ALvl Chemistry/2017/s/TZ 1/Paper 4/Q# 3/www.SmashingScience.org

3(a)(i)	A	1
3(a)(ii)	H	1
3(a)(iii)	G	1
3(a)(iv)	B	1
3(a)(v)	F	1

Q# 198/ ALvl Chemistry/2017/m/TZ 2/Paper 4/Q# 1/www.SmashingScience.org

1(b)(i)	P ₄ + 5O ₂ → P ₄ O ₁₀ / 2P ₂ O ₅	1
1(b)(ii)	any 2 from: - yellow / green colour (of chlorine gas) disappears - white flame - white solid - solid melts	2
1(b)(iii)	phosphoric(V) acid	1

Q# 199/ ALvl Chemistry/2016/s/TZ 1/Paper 4/Q# 2/www.SmashingScience.org

(ii)	$\frac{(A \times 0.56) + (86 \times 9.86) + (87 \times 7.00) + (88 \times 82.58)}{100} = 87.71$	[1]	[2]
	A = 84	[1]	
			[16]
2 (a)	D = Ga G = Se	[1]	[1]
(b) (i)	D₂O₃ + 6HCl → 2DCl₃ + 3H₂O M1 = species; M2 = balancing	[1]	[2]

(ii)	$\text{D}_2\text{O}_3 + 2\text{NaOH} + 7\text{H}_2\text{O} \rightarrow 2\text{NaD}(\text{OH})_4(\text{H}_2\text{O})_2$ OR $\text{D}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaD}(\text{OH})_4$ OR $\text{D}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaDO}_2 + \text{H}_2\text{O}$ OR $\text{D}_2\text{O}_3 + 2\text{OH}^- + 7\text{H}_2\text{O} \rightarrow 2[\text{D}(\text{OH})_4(\text{H}_2\text{O})_2]^-$ OR $\text{D}_2\text{O}_3 + 2\text{OH}^- + 3\text{H}_2\text{O} \rightarrow 2[\text{D}(\text{OH})_4]^-$ OR $\text{D}_2\text{O}_3 + 2\text{OH}^- \rightarrow 2\text{DO}_2^- + \text{H}_2\text{O}$ M1 = species; M2 = balancing	[1] [1]	[2]
(c)	giant ionic / ionic lattice	[1]	[1]
(d)	$\text{GO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{GO}_3$	[1]	[1]

Q# 200/ ALvl Chemistry/2015/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

(c) (ii)	Covalent AND simple / molecular	[1]	
	low melting point / reaction with water	[1]	[2]
(iii)	$\text{TeCl}_4 + 3\text{H}_2\text{O} \rightarrow \text{H}_2\text{TeO}_3 + 4\text{HCl}$ OR $\text{TeCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{TeO}_2 + 4\text{HCl}$	[1]	[1]
(d) (i)	Yellow / orange flame White fumes / solid Yellow / green gas disappears	[1] [1] [1]	[max 2]
(ii)	NaCl giant / lattice AND ionic SiCl ₄ simple / molecular AND covalent For NaCl large difference in electronegativity (of sodium / Na and chlorine / Cl / Cl ₂) (indicates electron transfer / ions) For SiCl ₄ smaller difference (indicates sharing / covalency) with (weak) van der Waals' / IM forces (between molecules) ora	[1] [1] [1] [1]	[4]

Q# 201/ ALvl Chemistry/2014/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

(d) (i)	$\text{NaCl} (+ \text{aq}) \rightarrow \text{Na}^+ + \text{Cl}^-$ $\text{NaCl} + \text{H}_2\text{O} \rightarrow \text{Na}^+ + \text{Cl}^- + \text{H}_2\text{O}$ $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$ $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{Si}(\text{OH})_4 + 4\text{HCl}$ $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{SiO}_2 \cdot 2\text{H}_2\text{O} + 4\text{HCl}$ Allow correct equation with other molar amounts of water	1 1	2
(ii)	NaCl is ionic AND giant / lattice NaCl dissolves / does not react SiCl ₄ is <u>covalent</u> AND molecular / simple SiCl ₄ is hydrolysed / reacts	1 1 1 1	4

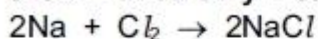


(b) sodium

burns with a yellow or orange flame or
forms a white solid

allow – once only – colour of chlorine disappears

(1)



(1)

phosphorus

burns with a white or yellow flame or

colour of chlorine disappears – if not given for Na – or

for PCl_5 forms a white or pale yellow solid

for PCl_3 forms a colourless liquid

(1)



or



equation must refer to compound described

(1) [4]

(d) MgCl_2 6.5 to 6.9

(1)

SiCl_4 0 to 3

(1)

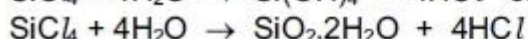
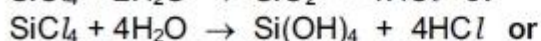
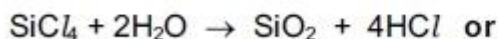
MgCl_2 dissolves without reaction or

slight or partial hydrolysis occurs

(1)

SiCl_4 reacts with water or
hydrolysis occurs

(1)



(1) [5]

3 (a) penalise (–1) for names of elements

(i) Na or K or Li

(1)

(ii) S or C or N or P

(1)

(iii) K

(1)

(iv) C

(1)

(v) Cl

(1)

(vi) Al or Si

(1) [6]



- (b) (i) Al_2O_3 or SiO_2 (1)
- (ii) Na_2O (1)
- (iii) P_2O_3 or P_4O_6 and P_2O_5 or P_4O_{10} or SO_2 and SO_3 (1+1)
- (iv) Al_2O_3 (1) [5]

Q# 204/ ALvl Chemistry/2012/w/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

- 1 (a) ZnCO_3 Zn(OH)_2 ZnO
not Zn or other compounds of Zn (any 2) [2]

Q# 205/ ALvl Chemistry/2012/s/TZ 1/Paper 4/Q# 1/www.SmashingScience.org

1 (a)

Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2	Cl_2O_7
alkaline	basic	amphoteric	acidic	acidic	acidic	acidic

- Na_2O is alkaline – allow basic (1)
- MgO is basic – allow alkaline (1)
- Al_2O_3 is amphoteric (1)
- SiO_2 , P_4O_{10} , and SO_2 are all acidic (1) [4]

- (b) any two from:
sodium, phosphorus, sulfur and chlorine
two names required (1) [1]

- (c) (i) any three from:
floats
vigorous/violent reaction occurs
melts/forms a sphere
moves
disappears – allow dissolves
effervescence/gas produced (any 3)

- (ii) $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \frac{1}{2}\text{H}_2$
or
 $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ (1) [4]



(d) (i) and (ii)

element	Na	Mg	Al	Si	P	S
conductivity	high	high	—	moderate	low	low
melting point	low	high	—	high	low	low

(1)

(1)

(1)

(1)

(1)

one mark for each correct column

[5]

(e) germanium/Ge

(1) [1]

3 (a) Accept only symbols.

(i) S or S₈ (1)(ii) K or K⁺ (1)

(iii) Na – allow K or Li (1)

(iv) Cl or Br or F (1)

(v) Mg or Ca or Li
allow Ni, Cu, or Zn (1)

[5]

(b) Accept only formulae.

(i) F₂O (1)(ii) SO₂ and SO₃
or P₂O₃/P₄O₆ and P₂O₅/P₄O₁₀
or any two from N₂O₃, NO₂/N₂O₄, N₂O₅
or any two from Cl₂O, ClO₂, ClO₃, Cl₂O₇ (1+1)

[3]

(c) chlorine is a strong/powerful oxidising agent

(1)

[1]

(e) covalent/not ionic

(1)

simple molecular or

mention of weak intermolecular forces or

weak van der Waals's forces between molecules

(1)

[2]

[Total: 14 max]

(b) (i) pass chlorine gas
over heated aluminium

(1)

(1)

(ii) aluminium glows
white/yellow solid formed
chlorine colour disappears/fades

(1)

(1)

(1) (any 2)

